BANDARU NIHANTH SAI

Site Reliability / DevOps Engineer | AWS | Kubernetes | Terraform | Observability
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PROFESSIONAL SUMMARY

DevOps and Cloud Engineer with hands-on experience in AWS, Ansible Automation Platform (AAP), Kubernetes/EKS, Terraform, Docker, CI/CD, and Prometheus/Grafana. Experienced in developing network automation workflows, provisioning cloud infrastructure, deploying containerized applications, and troubleshooting infrastructure and monitoring issues. Currently working as an Analyst at HCLTech.

TECHNICAL SKILLS

Cloud: AWS (EC2, S3, VPC, IAM, ALB, RDS, ECR, EKS)
Infrastructure as Code: Terraform
Containers & Orchestration: Docker, Kubernetes, Amazon EKS, Helm
CI/CD & Source Control: Git, GitHub, GitHub Actions, Jenkins
Automation & Scripting: Ansible, Ansible Automation Platform (AAP), Python, Bash, PowerShell
Monitoring & Observability: Prometheus, Grafana, Pushgateway
Systems & Networking: Linux, Ubuntu, VPC networking, load balancing

PROFESSIONAL EXPERIENCE

HCLTech — Analyst | December 2024 – Present | India
- Developed network automation workflows using Ansible Automation Platform (AAP) to automate operational tasks across network infrastructure.
- Implemented Ansible playbooks for network device onboarding, inventory management, connectivity validation, and compliance checks.
- Configured and executed automation jobs through AAP to streamline repetitive operational activities and improve execution consistency.
- Worked with Prometheus, Grafana, and Pushgateway to monitor and visualize metrics generated by automation workflows.
- Contributed to Grafana dashboards for tracking network device inventory, onboarding status, compliance, and automation outcomes.
- Troubleshot Ansible playbooks, automation execution failures, connectivity issues, and infrastructure-related problems.
- Collaborated with project teams to analyze requirements, validate automation workflows, and identify opportunities to improve operational efficiency.

PROJECTS

AWS 3-Tier Application | AWS | EC2 | VPC | ALB | RDS MySQL | Linux
- Designed and deployed separate frontend, backend, and database tiers on AWS.
- Built a multi-AZ VPC with public and private subnets; placed frontend workloads behind a public ALB and backend workloads behind an internal ALB.
- Configured Amazon RDS MySQL in private networking and troubleshot connectivity through routing, security groups, ports, and application settings.

Kubernetes Application Deployment on Amazon EKS | AWS EKS | Kubernetes | Docker | Helm | ALB | IAM/IRSA | EBS CSI
- Created an EKS cluster and deployed workloads using Deployments, Services, Ingress, ConfigMaps, service accounts, and storage resources.
- Configured OIDC/IRSA permissions, AWS Load Balancer Controller, ALB Ingress, and EBS CSI for persistent storage.
- Troubleshot 503 responses, unhealthy targets, and Availability Zone-related load-balancer configuration issues.

Infrastructure Automation with Terraform | Terraform | AWS | VPC | EC2 | EKS | IAM
- Automated repeatable AWS infrastructure provisioning with Terraform, including VPCs, subnets, security groups, IAM components, and EKS-relatedresources.
- Used variables, outputs, dependencies, and reusable design patterns to improve consistency across environment creation and teardown.

CI/CD Pipeline for Containerized Applications | GitHub Actions | Docker | AWS ECR | Amazon EKS
- Built CI/CD workflows to build container images and publish them to Amazon ECR.
- Practiced automated deployment workflows for EKS, including image updates and Kubernetes rollout management.
- Worked with code-quality and security-scanning concepts using SonarQube and Trivy.

Prometheus & Grafana Observability | Prometheus | Grafana | Pushgateway | PromQL
- Worked with metrics, labels, counters, gauges, and PromQL functions including rate() and increase().
- Created Grafana dashboards for infrastructure and automation-related metrics and explored alerting for resource utilization.
- Used Pushgateway for short-lived automation jobs; Prometheus scrapes standard targets directly for ongoing monitoring.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03) — Amazon Web Services (AWS) | 2026

EDUCATION

Bachelor of Engineering in Computer Science & Engineering | R.M.K. Engineering College, Chennai | 2020–2024